COMPREHENSIVE ANALYTICAL REPORT

# PMARIS Asset Reliability: where the next EGP of maintenance should go

A full analysis of 5,000 thirty-day asset-health observations across six industrial sites in four industries, September 2025 to August 2026, prepared for the Head of Reliability & Maintenance.

**THE ANSWER**

Put the next EGP on four controllable levers and on a wider escalation list. The top 10% of observation windows carry **44.7%** of revenue loss; that concentration tracks maintenance strategy, spare-parts availability, crew skill and PM discipline, each worth roughly 2x on downtime. The executive priority flag misses **829** of the **960** high-risk windows.

| 1.88B | 11.0 h | 19.2% | 46.7% | 829 |
|---|---|---|---|---|
| EGP revenue loss, all windows | mean unplanned downtime / window | windows high-risk (FP ≥ 50%) | of loss in those windows | high-risk, only P3/P4 |

| Prepared for | Prepared by | Method | Status |
|---|---|---|---|
| Head of Reliability & Maintenance | AIB902 Group 9230, Team 1 | AIB902 Agentic Analytics Playbook v4.0 | Final, 11 October 2026 |

*Companion deliverables: the live six-page dashboard (03_Dashboard), the 15-slide Executive Presentation, the 30-slide full deck and the 10-page assessment. All are generated from the same analysis output, 02_Analysis/findings.json, so their figures agree.*

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How to read this report. Sections 1 and 22 are the decision; sections 3 to 7 establish that the data can be trusted and how far; sections 8 to 19 are the evidence, one question per section; sections 20 and 21 are the statistical record and the limits. Every number is measured from the file unless it is marked ESTIMATED.

SECTION 1

# Executive summary

The six sites lost **EGP 1.88B** of revenue across 5,000 observation windows, to **54,938** hours of unplanned downtime, against **EGP 126.4M** of maintenance spend. Three arguments support the answer, each sufficient on its own.

## 1. Loss is concentrated where risk is high

The top 10% of windows carry 44.7% of revenue loss (Gini 0.63); the 960 windows with failure probability of 50% or more carry 46.7% of loss and 45.0% of downtime. Windows under health 70 lose 9x those above 90.

## 2. The concentration tracks levers management controls

Reactive maintenance: 19.0 h of downtime per window against 8.4 h for predictive (2.3x), at 1.6x the maintenance cost. Parts shortage: 20.8 h against 7.2 h available (2.9x). Basic crews 1.8x expert. PM compliance below 80%: 2.2x. Every split survives a Bonferroni-corrected test with non-overlapping 95% intervals.

## 3. The executive escalation flag does not see most of it

Only 134 windows are P1 or P2. 829 of 960 high-risk windows sit at P3 or P4 with EGP 723.3M of exposure; P1/P2 cover 26.5% of the loss in the worst 10% of windows.

## What this report recommends

| # | Action | Why |
|---|---|---|
| 1 | Move reactive windows onto a planned strategy, critical assets first | 2.3x downtime, 56% emergency WOs |
| 2 | Fund spare-parts availability for shortage-prone classes | 2.9x downtime; largest categorical effect |
| 3 | Manage to PM ≥ 80%, maintenance < 60 days, backlog < 14 days | each threshold splits downtime about 2x |
| 4 | Pair basic crews with experts on high-risk work | 1.8x downtime gap |
| 5 | Add a second escalation trigger after a capacity check | +433 windows, EGP 434.3M exposure |
| 6 | Pilot before scaling | turns association into a causal estimate |
| 7 | Fix Asset_ID in the next export | unlocks per-asset planning |

SECTION 2

# Business context and the decision

The scope was agreed with the stakeholder before any dashboard or report was built, through the Playbook's scoping interview (Stage 4). The resulting Dashboard Scope Brief is the contract for every deliverable.

| Element | Agreed scope |
|---|---|
| Audience | Head of Reliability & Maintenance, accountable across all six sites |
| Decision | Where to put the next EGP: which controllable lever to fund, and which windows to escalate |
| KPIs | Mean revenue loss per window; mean unplanned downtime; high-risk share and the loss it carries; escalation gap |
| Refresh | Live (watcher re-computes within about 0.3 s of a save; verified under 1 s end to end) |
| Headline | The next EGP belongs on the controllable levers and on the escalation list |
| Out of scope | Per-asset replacement or remaining-life planning (Asset_ID is not a stable key, Section 6) |

## The operating footprint

| Site | Industry | Windows | Mean loss | Mean downtime | High-risk |
|---|---|---|---|---|---|
| Cement Plant A | Cement Manufacturing | 1,009 | EGP 433.5k | 12.1 h | 22.7% |
| Drilling Site East | Oil & Gas Drilling | 877 | EGP 404.9k | 11.9 h | 24.6% |
| Cement Plant B | Cement Manufacturing | 830 | EGP 422.6k | 12.7 h | 24.3% |
| Lab Equipment Facility | Laboratory Equipment Manufacturing | 807 | EGP 291.5k | 8.6 h | 11.4% |
| Diagnostics Service Center | In-Vitro Diagnostics / Medical | 756 | EGP 277.4k | 8.4 h | 9.4% |
| Drilling Site West | Oil & Gas Drilling | 721 | EGP 402.6k | 11.8 h | 20.8% |

## Why this decision matters

Revenue lost to failure is 14.9x what the sites spent on maintenance in the same windows (EGP 1.88B against EGP 126.4M). A maintenance budget allocated in proportion to where loss actually occurs, rather than evenly or by site, is the largest single opportunity in the file.

SECTION 3

# Methodology

The file was taken through the AIB902 Agentic Analytics Playbook (v4.0) in a fixed order. Each stage's output is the next stage's input, and two gates (an approved scope brief, and a timed live-refresh test) were passed before anything was delivered.

| Stage | What was done | Output |
|---|---|---|
| 0 Intake | Format, grain, time dimension, dimensions, measures, scale | Path A: one flat table with a date column |
| 1 Cleaning | Types, missing, duplicates, outliers, constant and rare levels, bounds | Zero defects; nothing dropped (Section 5) |
| 1b Meaning | Entity-key stability, derived-column detection, flag banding | Broken Asset_ID; derived scores (Section 6) |
| 3 Insights | Totals, rates, concentration, per-dimension profiles, trend, correlations | Sections 7 to 19 |
| 3b Significance | ANOVA + Kruskal-Wallis, Bonferroni, $\eta^2$, bootstrap intervals | 26 of 33 differences real (Section 20) |
| 4 Scoping | Stakeholder interview: audience, decision, KPIs, refresh, pages | Approved scope brief (Section 2) |
| 5 Build + verify | Live dashboard, report, decks; headless render and timed edits | 376 ms mean refresh (Appendix) |

## Statistical tests and why each was chosen

| Question | Test | Rule applied |
|---|---|---|
| Do groups differ on a measure? | One-way ANOVA (F), with $\eta^2$ | $\eta^2 \geq 0.01$ small, $\geq 0.06$ medium, $\geq 0.14$ large |
| Is that robust to skew and outliers? | Kruskal-Wallis H | Must agree with ANOVA to count |
| Too many tests? | Bonferroni correction | 0.05 / 33 = 0.0015 |
| Is a ranking reproducible? | Bootstrap, 2,000 resamples | Non-overlapping 95% intervals |
| Is loss concentrated? | Lorenz curve, Gini, top-decile share | Top 10% share reported |
| Is there a trend or momentum? | Linear slope, lag-1 autocorrelation | Slope as % of mean per month |
| Does the flag catch the risk? | Cross-tab of priority vs outcome deciles | Share of top-decile loss covered |

## Why effect size, not p-value, drives the wording

With 5,000 rows almost any difference is statistically detectable. A p-value says a difference is not chance; it does not say the difference is worth a budget line. This report therefore calls a difference a lever only when it survives both tests and carries a non-trivial effect size, and it describes real-but-small effects as small.

SECTION 4

# Data dictionary

All 60 columns, with their role in the analysis and the values they actually take. 'Derived' marks a composite score or flag computed from other columns; these are reported but never treated as independent evidence.

| Column | Role | Values in this file |
|---|---|---|
| Observation ID | Identifier | 5,000 distinct |
| Asset ID | Identifier | 500 distinct |
| Observation Date | Date | 2025-09-01 to 2026-08-31 |
| Site | Dimension | 6 levels: Cement Plant A, Drilling Site East, Cement Plant B, Lab Equipment Facility +2 more |
| Industry Context | Dimension | 4 levels: Cement Manufacturing, Oil & Gas Drilling, Laboratory Equipment Manufacturing, In-Vitro Diagnostics / Medical |
| Asset Type | Dimension | 25 levels: Compressor, Pump, Crusher, Shale Shaker +21 more |
| Asset Class | Dimension | 10 levels: Rotating Equipment, Rotating Heavy Equipment, Drilling Equipment, Utility Equipment +6 more |
| Manufacturer Group | Dimension | 6 levels: OEM-A, OEM-B, OEM-C, OEM-D +2 more |
| Criticality Class | Dimension | 4 levels: High, Medium, Critical, Low |
| Maintenance Strategy | Dimension | 4 levels: Preventive, Predictive, Reactive, Condition-Based |
| Failure Mode Category | Dimension | 6 levels: General Mechanical, Bearing / Vibration, Lubrication, Control / Sensor +2 more |
| Work Order Type | Dimension | 4 levels: Corrective Planned, Corrective Emergency, Preventive, Inspection |
| Spare Parts Status | Dimension | 3 levels: Available, Limited, Shortage |
| Technician Skill Level | Dimension | 4 levels: Intermediate, Advanced, Expert, Basic |
| Operating Shift | Dimension | 3 levels: Day, Evening, Night |
| Environment Severity | Dimension | 4 levels: Harsh, Moderate, Severe, Controlled |
| Sensor Health Status | Dimension | 3 levels: Healthy, Watch, Degraded |
| Inspection Status | Dimension | 3 levels: Pass, Attention, Fail |
| Asset Age Years | Measure | 0.3 to 22.0 (median 5.9) |
| Operating Hours 30D | Measure | 86.6 to 720.0 (median 540.6) |
| Utilization Rate Pct | Measure | 22.2 to 99.0 (median 75.2) |
| Load Factor Pct | Measure | 15.0 to 110.0 (median 77.3) |
| Temperature C | Measure | 40.4 to 84.4 (median 63.6) |
| Vibration mm s | Measure | 1.8 to 12.0 (median 6.7) |
| Pressure Bar | Measure | 2.8 to 25.3 (median 12.8) |
| Lubricant Quality Index | Measure | 20.0 to 100.0 (median 54.9) |
| Energy Consumption kWh | Measure | 51.2 to 441.0 (median 248.9) |
| Starts Stops 30D | Measure | 5.0 to 40.0 (median 19.0) |
| Failure Count 12M | Measure | 0.0 to 10.0 (median 1.0) |
| Days Since Last Maintenance | Measure | 1.0 to 180.0 (median 47.6) |
| Preventive Maintenance Compliance Pct | Measure | 20.7 to 100.0 (median 83.2) |
| Maintenance Backlog Days | Measure | 0.3 to 65.7 (median 12.4) |
| Mean Time Between Failures Hours | Measure | 331.4 to 3,282.2 (median 1,182.0) |
| Mean Time To Repair Hours | Measure | 0.5 to 18.0 (median 5.5) |
| Unplanned Downtime Hours 30D | Measure | 0.0 to 120.0 (median 6.5) |
| Planned Downtime Hours 30D | Measure | 0.0 to 17.5 (median 9.4) |
| Maintenance Cost EGP 30D | Measure | 1,200.0 to 186,251.2 (median 19,433.6) |
| Spare Parts Cost EGP 30D | Measure | 0.0 to 18,215.2 (median 2,881.7) |
| Labor Hours 30D | Measure | 1.0 to 159.0 (median 25.6) |
| Spare Parts Lead Time Days | Measure | 1.0 to 36.2 (median 11.9) |
| Sensor Anomaly Count 30D | Measure | 0.0 to 15.0 (median 3.0) |

| Column | Role | Values in this file |
|---|---|---|
| Alarm Count 30D | Measure | 0.0 to 20.0 (median 6.0) |
| Inspection Defect Count | Measure | 0.0 to 8.0 (median 1.0) |
| Production Loss Units | Measure | 0.0 to 15,116.0 (median 556.4) |
| Revenue Loss EGP 30D | Measure | 0.0 to 6,000,000.0 (median 181,970.7) |
| Replacement Cost EGP | Measure | 641,383.9 to 3,706,018.8 (median 1,441,452.1) |
| Safety Incident Count 12M | Measure | 0.0 to 4.0 (median 0.0) |
| Quality Defect Rate Pct | Measure | 0.1 to 8.1 (median 1.5) |
| Asset Health Index | Derived score | 52.5 to 100.0 (median 86.2) |
| Failure Probability Pct | Derived score | 1.0 to 95.0 (median 24.5) |
| Reliability Risk Score | Derived score | 0.6 to 73.2 (median 17.8) |
| Maintenance Priority Index | Derived score | 9.5 to 80.0 (median 34.6) |
| Downtime Risk Score | Derived score | 3.6 to 82.0 (median 25.4) |
| Remaining Useful Life Months | Derived score | 1.0 to 31.6 (median 19.4) |
| Maintenance Cost Efficiency Index | Derived score | 0.0 to 100.0 (median 100.0) |
| Operational Criticality Score | Derived score | 26.7 to 99.7 (median 69.0) |
| Preventive Maintenance Effectiveness Index | Derived score | 5.7 to 83.2 (median 58.5) |
| Replacement Priority Score | Derived score | 8.7 to 69.6 (median 31.9) |
| Financial Exposure At Risk EGP | Derived score | 10,078.5 to 6,221,459.8 (median 240,239.1) |
| Executive Intervention Priority | Derived flag | 4 levels: P4 - Monitor, P3 - Planned, P2 - High, P1 - Immediate |

SECTION 5

# Data quality assessment

Stage 1 ran the Playbook's universal checklist. The file passed with zero defects; nothing was dropped or imputed, and every judgement call is recorded below with its reason.

| Check | Result | Decision | Reason |
|---|---|---|---|
| Missing cells | 0 of 300,000 | **None needed** | - |
| Exact duplicates / duplicate IDs | 0 / 0 | **None needed** | - |
| Numeric type errors / bad dates | 0 / 0 | **None needed** | - |
| Revenue loss beyond 3 SD | 113 rows | Kept | A real heavy tail; removing it would hide the concentration |
| Downtime beyond 3 SD | 102 rows | Kept | Same reason |
| Load factor above 100% | 452 rows | Kept | Overload is physically possible |
| Failure modes under 1% of rows | 3 levels, 27 rows | Pooled as 'Other' | Too few rows per level to test separately |
| P1 - Immediate | 2 rows | Kept, combined with P2 as 'escalated' | Too rare to analyse alone |
| Rows per month | 357 to 443 | Kept | No gap in the 12-month range |

## Columns piled at a bound

Columns with at least 5% of rows at their minimum or maximum. A pile at a maximum usually means a capped scale, so values there are read as 'at or above the cap', not as measured.

| Column | Min | % at min | Max | % at max | Interpretation |
|---|---|---|---|---|---|
| Utilization Rate Pct | 22 | 0.0% | 99 | 5.6% | Capped index |
| Failure Count 12M | 0 | 34.5% | 10 | 0.0% | Many zero windows (no event) |
| Preventive Maintenance Compliance Pct | 21 | 0.0% | 100 | 15.8% | Capped index |
| Unplanned Downtime Hours 30D | 0 | 9.2% | 120 | 0.1% | Many zero windows (no event) |
| Spare Parts Cost EGP 30D | 0 | 17.2% | 18,215 | 0.0% | Many zero windows (no event) |
| Inspection Defect Count | 0 | 25.1% | 8 | 0.0% | Many zero windows (no event) |
| Production Loss Units | 0 | 13.1% | 15,116 | 0.0% | Many zero windows (no event) |
| Revenue Loss EGP 30D | 0 | 13.1% | 6,000,000 | 0.0% | Many zero windows (no event) |
| Safety Incident Count 12M | 0 | 72.9% | 4 | 0.1% | Many zero windows (no event) |
| Maintenance Cost Efficiency Index | 0 | 0.7% | 100 | 53.2% | Capped index |

SECTION 6

# Do the columns mean what they say?

A clean bill of health is not fitness for purpose. Playbook Stage 1b checks that identifiers identify and that measures relate the way their names imply. One column fails; several are formulas of others.

## Asset_ID is not an asset key

| Attribute that should be fixed per asset | Asset IDs where it changes (of 500) |
|---|---|
| Site | 499 |
| Asset type | 498 |
| Criticality class | 498 |
| Manufacturer group | 498 |
| Maintenance strategy | 499 |
| Asset age runs backwards in date order | 497 |

**CONSEQUENCE**

Every row is treated as an independent 30-day observation window. No per-asset history, remaining-life tracking, replacement plan or churn view can be built from this export, and none is attempted.

## Derived columns

| Column | Evidence | Treatment |
|---|---|---|
| Financial exposure at risk | r = 0.999 with revenue loss | Never added to loss; used only for the watchlist ranking |
| Failure probability | r = -0.972 with health index | Used for the high-risk threshold; health reported in bands |
| Executive Intervention Priority | Exact bands of the Maintenance Priority Index: P4 9.5–40.0, P3 40.0–59.9, P2 60.0–74.0, P1 79.2+ | Treated as a formula that can be re-tuned (Section 18) |
| Risk, priority, RUL, efficiency scores | Inter-correlated at 0.7 to 0.99 | Excluded from the drivers analysis (Section 15) |

SECTION 7

# Descriptive statistics

The core measures per 30-day window. Means sit well above medians for downtime, loss and cost: the distributions are right-skewed, which is why the rank-based test is required alongside ANOVA and why concentration matters.

| Measure | Mean | Median | Std dev | 90th pct | Max |
|---|---|---|---|---|---|
| Unplanned Downtime Hours 30D | 10.99 | 6.55 | 13.84 | 27.56 | 120.00 |
| Planned Downtime Hours 30D | 8.99 | 9.38 | 2.84 | 12.25 | 17.49 |
| Revenue Loss EGP 30D | 375,718 | 181,971 | 544,681 | 980,079 | 6,000,000 |
| Production Loss Units | 1,079 | 556 | 1,515 | 2,785 | 15,116 |
| Maintenance Cost EGP 30D | 25,289 | 19,434 | 20,797 | 50,674 | 186,251 |
| Spare Parts Cost EGP 30D | 3,383 | 2,882 | 3,044 | 7,550 | 18,215 |
| Labor Hours 30D | 29.82 | 25.60 | 16.55 | 49.40 | 159.00 |
| Failure Probability Pct | 30.35 | 24.48 | 21.74 | 64.22 | 95.00 |
| Asset Health Index | 84.57 | 86.15 | 9.23 | 95.28 | 100.00 |
| Mean Time Between Failures Hours | 1,274.09 | 1,181.95 | 518.33 | 2,008.30 | 3,282.20 |
| Mean Time To Repair Hours | 5.67 | 5.47 | 1.92 | 8.21 | 18.03 |
| Preventive Maintenance Compliance Pct | 80.81 | 83.18 | 16.30 | 100.00 | 100.00 |
| Maintenance Backlog Days | 13.79 | 12.40 | 7.65 | 23.90 | 65.70 |
| Days Since Last Maintenance | 52.24 | 47.60 | 28.07 | 90.10 | 180.00 |
| Vibration mm s | 6.69 | 6.69 | 1.29 | 8.34 | 11.96 |
| Temperature C | 63.56 | 63.59 | 6.02 | 71.28 | 84.36 |
| Lubricant Quality Index | 53.50 | 54.88 | 16.29 | 73.67 | 100.00 |
| Asset Age Years | 6.61 | 5.86 | 3.73 | 11.68 | 22.00 |
| Safety Incident Count 12M | 0.33 | 0.00 | 0.60 | 1.00 | 4.00 |

## Category mix

| Dimension | Levels and share of windows |
|---|---|
| Maintenance strategy | Preventive 41%, Predictive 23%, Reactive 18%, Condition-Based 18% |
| Spare-parts status | Available 48%, Limited 44%, Shortage 9% |
| Criticality | High 36%, Medium 34%, Critical 18%, Low 12% |
| Technician skill | Intermediate 38%, Advanced 36%, Expert 14%, Basic 12% |
| Environment | Harsh 37%, Moderate 28%, Severe 21%, Controlled 14% |

SECTION 8

# Finding 1: loss is concentrated

A tenth of the windows carry 44.7% of revenue loss and a fifth carry 64.8%. Downtime is almost as concentrated: the top 10% of windows carry 39.7% of unplanned hours (Gini 0.58). 656 windows (13.1%) lost nothing at all.

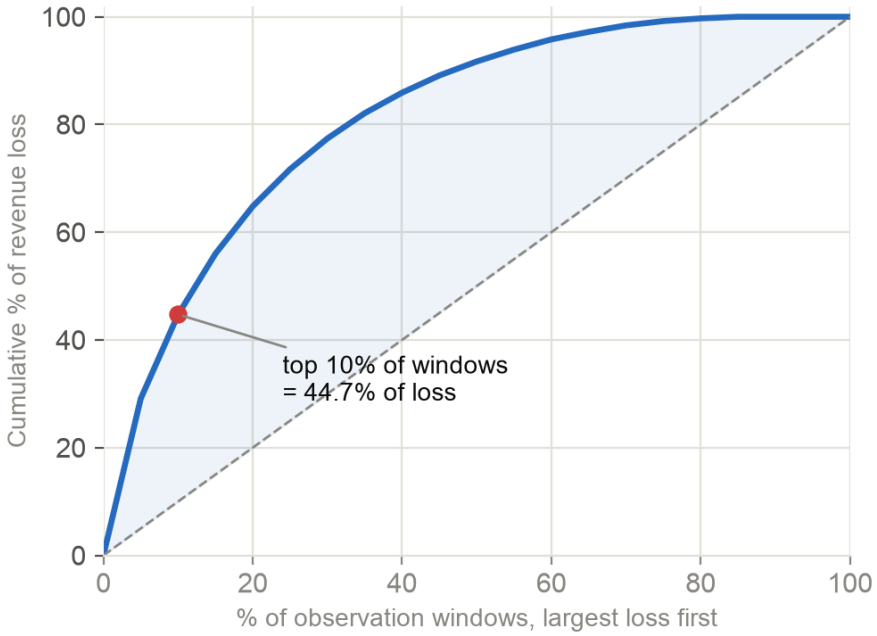



*Figure 1. Lorenz curve of revenue loss, windows sorted largest first. The dashed diagonal is an even spread.*

| Concentration measure | Revenue loss | Unplanned downtime |
|---|---|---|
| Share in the top 10% of windows | 44.7% | 39.7% |
| Gini coefficient | 0.631 | 0.583 |
| Share in high-risk windows (FP ≥ 50%) | 46.7% | 45.0% |

**SO WHAT**

Effort spread evenly across the fleet is mostly spent where little is lost. The 19.2% of windows that are high-risk are where 46.7% of the money goes.

SECTION 9

# Asset health, failure modes and risk

Loss rises steeply as health falls. Windows below a health index of 70 average EGP 1.1M of loss and 31.7 h of downtime, against EGP 125.4k and 3.9 h above 90. Each band down roughly doubles the loss, which is the case for intervening early.

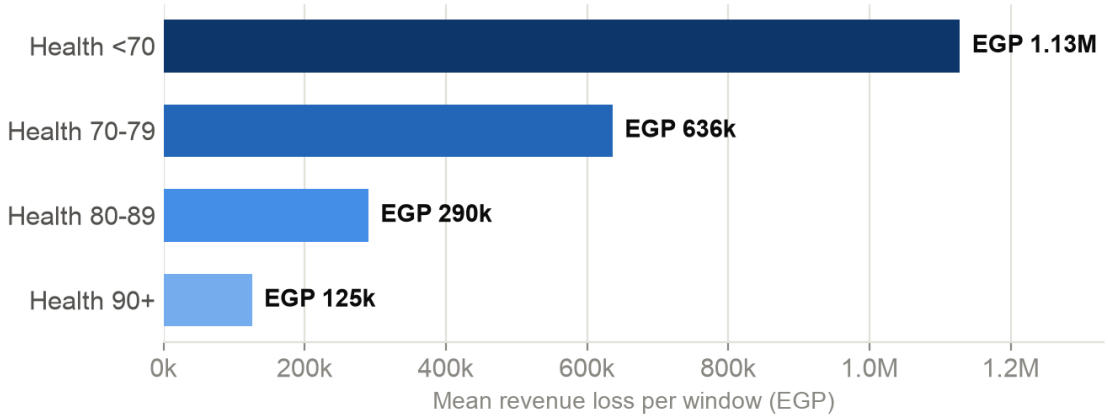



*Figure 2. Mean revenue loss per window by Asset Health Index band.*

Failure mode is the strongest single categorical signal in the file ($\eta^2$ 0.112 on downtime). Bearing and vibration failures average EGP 628.1k per window, 2.6x general mechanical failures. Failure mode describes what went wrong rather than a lever, but it tells the crews where condition monitoring (vibration, lubrication) pays.

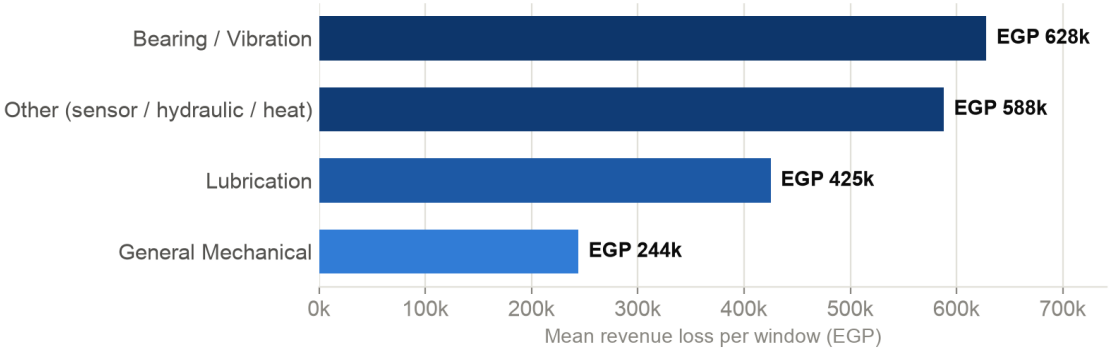



*Figure 3. Mean revenue loss per window by failure mode. 'Other' pools 27 rows across three rare modes.*

SECTION 10

## Time: a flat year

Mean revenue loss per window moves within a narrow band all year. The fitted slope is +0.60% of the monthly mean per month, and the lag-1 autocorrelation is -0.17: a bad month does not predict the next. There is no trend to extrapolate and no season to schedule maintenance around.

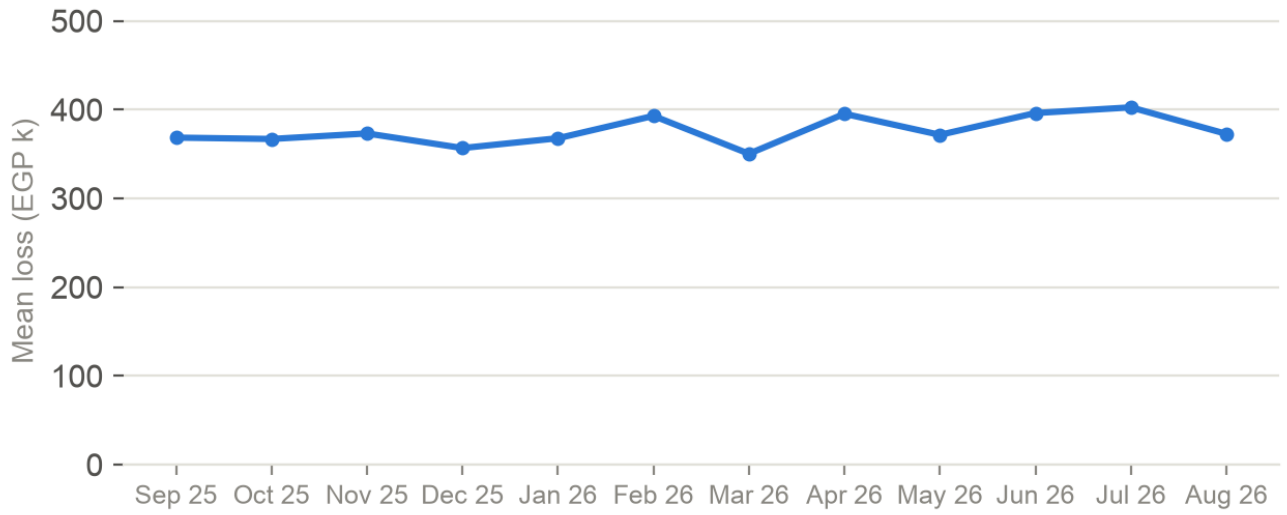



*Figure 4. Mean revenue loss per window by month, September 2025 to August 2026.*

| Month | Windows | Mean loss | Mean downtime | Failure prob. | Health |
|---|---|---|---|---|---|
| Sep 2025 | 441 | EGP 368.5k | 10.6 h | 29.9% | 84.8 |
| Oct 2025 | 421 | EGP 366.9k | 10.6 h | 30.0% | 84.9 |
| Nov 2025 | 396 | EGP 373.2k | 10.8 h | 30.1% | 84.7 |
| Dec 2025 | 443 | EGP 356.7k | 11.2 h | 30.8% | 84.4 |
| Jan 2026 | 425 | EGP 367.8k | 10.5 h | 31.2% | 84.3 |
| Feb 2026 | 357 | EGP 392.9k | 11.2 h | 29.8% | 84.8 |
| Mar 2026 | 428 | EGP 350.3k | 10.2 h | 29.5% | 84.9 |
| Apr 2026 | 386 | EGP 395.3k | 11.4 h | 30.4% | 84.5 |
| May 2026 | 433 | EGP 371.6k | 11.1 h | 29.8% | 84.6 |
| Jun 2026 | 420 | EGP 396.2k | 10.9 h | 30.5% | 84.7 |
| Jul 2026 | 424 | EGP 402.5k | 11.5 h | 31.1% | 84.1 |
| Aug 2026 | 426 | EGP 372.6k | 11.8 h | 31.1% | 84.3 |

**SO WHAT**

Plan maintenance by asset condition, not by calendar.

SECTION 11

# Lever 1: maintenance strategy

Reactive windows average 19.0 h of unplanned downtime; the three planned strategies sit between 8.4 and 9.5 h with overlapping intervals (η² 0.076, p = 4e-85). The step that matters is leaving reactive, not choosing between planned strategies. Reactive is also not the cheap option: it costs more to maintain, not less.

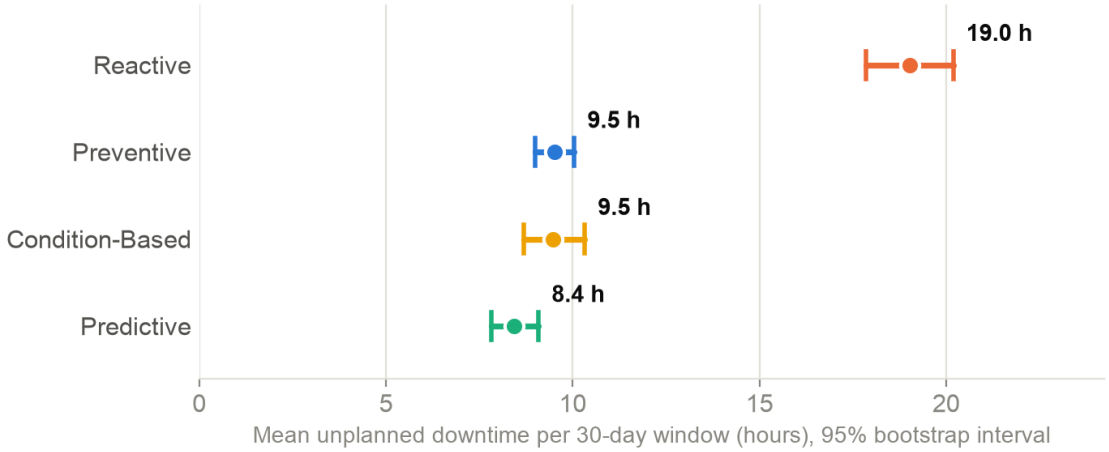



*Figure 5. Mean downtime by strategy with 95% bootstrap intervals (2,000 resamples).*

| Strategy | Windows | Downtime | Revenue loss | Maint. cost | Spare parts | Labour h | Emergency WOs |
|---|---|---|---|---|---|---|---|
| Reactive | 906 | 19.0 h | EGP 623.8k | EGP 34.7k | EGP 4.6k | 33.6 | 55.8% |
| Preventive | 2,051 | 9.5 h | EGP 334.0k | EGP 23.7k | EGP 3.2k | 29.4 | 25.6% |
| Condition-Based | 901 | 9.5 h | EGP 329.0k | EGP 23.4k | EGP 3.0k | 29.4 | 26.4% |
| Predictive | 1,142 | 8.4 h | EGP 290.6k | EGP 22.1k | EGP 3.1k | 28.0 | 23.0% |

**SO WHAT**

Reactive windows run 56% emergency work orders against 23% for predictive, and cost EGP 34.7k per window to maintain against EGP 22.1k.

SECTION 12

# Lever 2: spare-parts availability

A clean staircase: shortage 20.8 h, limited 13.2 h, available 7.2 h, with no interval touching the next (η² 0.091 on downtime, the largest among the controllable dimensions). Shortages also cost more on every line: parts bought at a premium, more labour, more maintenance.

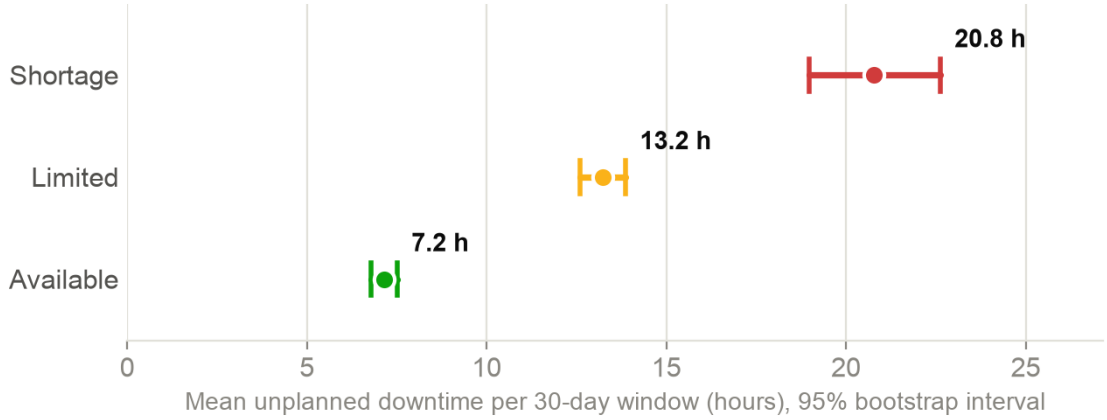



*Figure 6. Mean downtime by spare-parts status with 95% bootstrap intervals.*

| Parts status | Windows | Downtime | Revenue loss | Maint. cost | Spare parts | Labour h | High-risk |
|---|---|---|---|---|---|---|---|
| Shortage | 426 | 20.8 h | EGP 704.4k | EGP 39.1k | EGP 9.1k | 39.8 | 50.0% |
| Limited | 2,196 | 13.2 h | EGP 449.8k | EGP 28.4k | EGP 3.1k | 31.8 | 24.4% |
| Available | 2,378 | 7.2 h | EGP 248.5k | EGP 20.0k | EGP 2.6k | 26.2 | 8.9% |

**SO WHAT**

Shortage windows spend 3.5x as much on spare parts and 2.0x as much on maintenance as windows with stock available, while losing far more revenue. Holding stock is cheaper than not holding it.

SECTION 13

# Levers 3 and 4: crew skill and PM discipline

Basic crews average 14.8 h of downtime per window, intermediate 12.4 h, advanced 9.4 h and expert 8.0 h. 2,509 windows (50%) are worked by basic or intermediate crews.

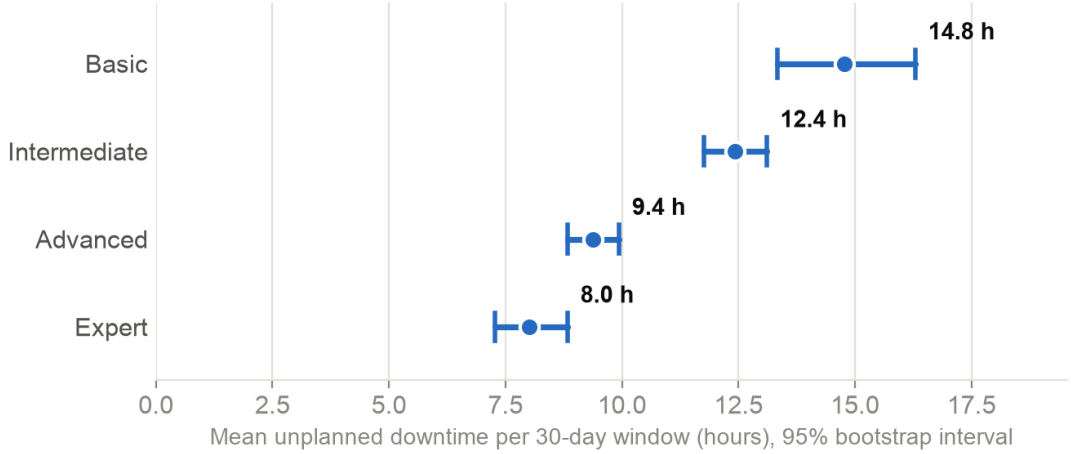



*Figure 7. Mean downtime by technician skill with 95% bootstrap intervals.*

PM compliance shows a dose-response: the least compliant decile averages 22.8 h, the most compliant 6.5 h, falling almost every step. A steady gradient across ten groups is stronger evidence than any single split.

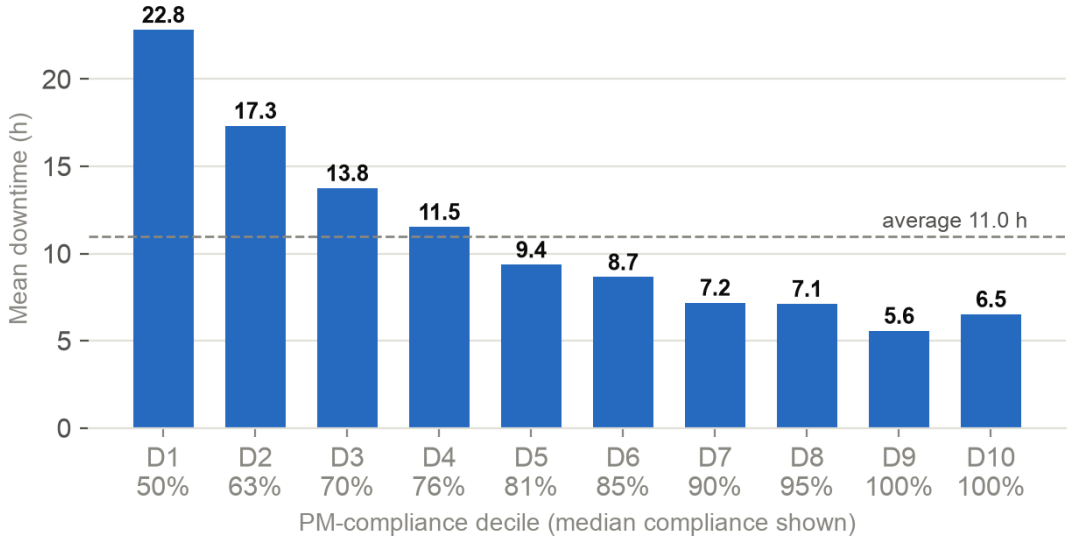



*Figure 8. Mean downtime by PM-compliance decile (median compliance under each bar).*

SECTION 14

# Operating thresholds and compounding

Each lever was split at a practical threshold the maintenance team can manage to. Every split roughly halves downtime and is significant far beyond the corrected threshold.

| Lever | Worse side | Windows | Downtime | Better side | Downtime | Ratio | p | $\eta^2$ |
|---|---|---|---|---|---|---|---|---|
| PM compliance | below 80% | 2,117 | 16.0 h | 80% or more | 7.3 h | 2.2x | 4e-113 | 0.097 |
| Days since maintenance | 60+ days | 1,636 | 15.7 h | under 60 days | 8.7 h | 1.8x | 4e-65 | 0.056 |
| Lubricant quality | LQI under 50 | 1,956 | 15.8 h | LQI 50+ | 7.9 h | 2.0x | 7e-90 | 0.078 |
| Maintenance backlog | backlog 14d+ | 2,124 | 15.5 h | backlog under 14d | 7.7 h | 2.0x | 1e-89 | 0.078 |

The levers compound. Reactive maintenance with a parts shortage averages 26.3 h of downtime; predictive with parts available averages 5.9 h, a 4.5x spread. Even with parts available, reactive runs 11.4 h against 5.9 h for predictive, so the two levers are not substitutes.

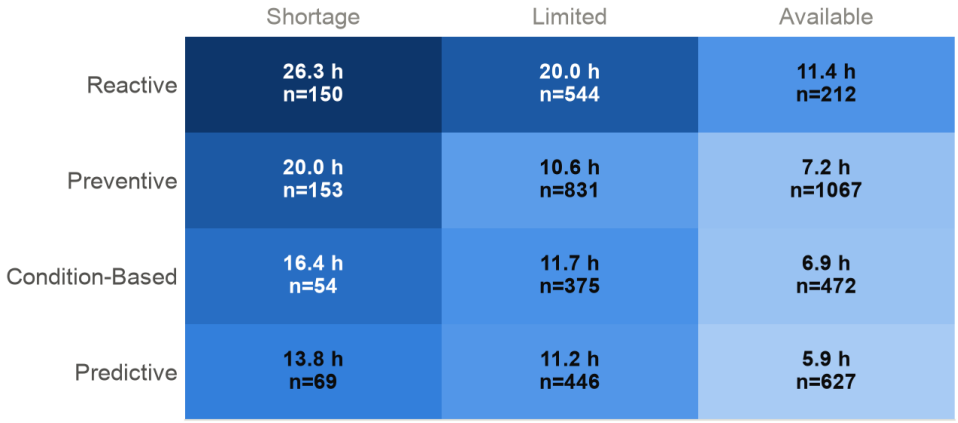



*Figure 9. Mean downtime by strategy × spare-parts status, with observation counts.*

**TARGETS**

Operating targets: PM compliance at or above 80%, maintenance within 60 days, lubricant quality above 50, backlog under 14 days.

SECTION 15

# Measured drivers of downtime

Correlation of each measured, non-derived input with unplanned downtime. Composite scores are excluded because they are formulas of these inputs and would 'explain' downtime by construction (Section 6).

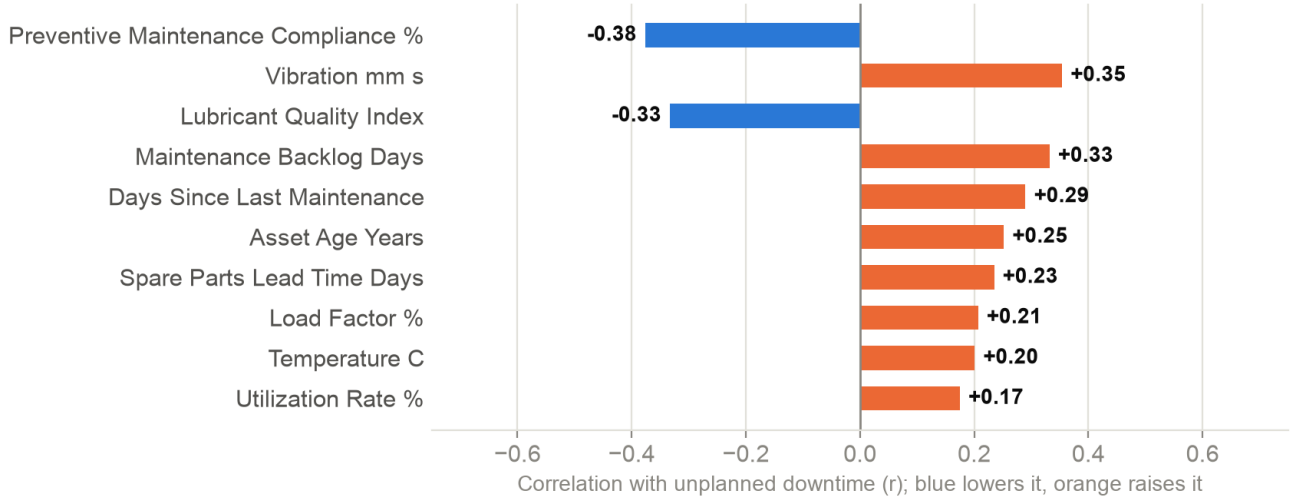



*Figure 10. Correlation with unplanned downtime. Blue inputs lower it; orange raise it.*

| Input | r with downtime | Reading |
|---|---|---|
| Preventive Maintenance Compliance Pct | -0.376 | Controllable |
| Vibration mm s | +0.354 | Condition / operating context |
| Lubricant Quality Index | -0.333 | Controllable |
| Maintenance Backlog Days | +0.332 | Controllable |
| Days Since Last Maintenance | +0.290 | Controllable |
| Asset Age Years | +0.252 | Condition / operating context |
| Spare Parts Lead Time Days | +0.235 | Controllable |
| Load Factor Pct | +0.207 | Condition / operating context |
| Temperature C | +0.201 | Condition / operating context |
| Utilization Rate Pct | +0.175 | Condition / operating context |

**SO WHAT**

The strongest measured inputs are ones maintenance controls: PM compliance, lubrication, backlog and time since the last service. Vibration is the best condition signal for early warning.

SECTION 16

# Sites, criticality and environment

Cement and drilling sites lose more per window than the lab and diagnostics sites, and the gap is real (p = 7e-13), but site explains only 1.3% of loss variance. Criticality matters more for money: critical windows average EGP 596.5k, 4.4x low-criticality ones (η² 0.064).

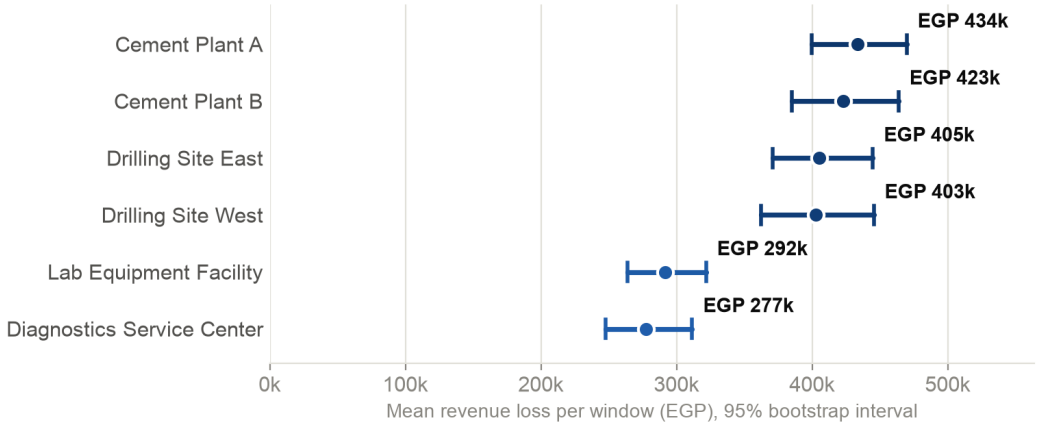



*Figure 11. Mean revenue loss by site with 95% bootstrap intervals.*

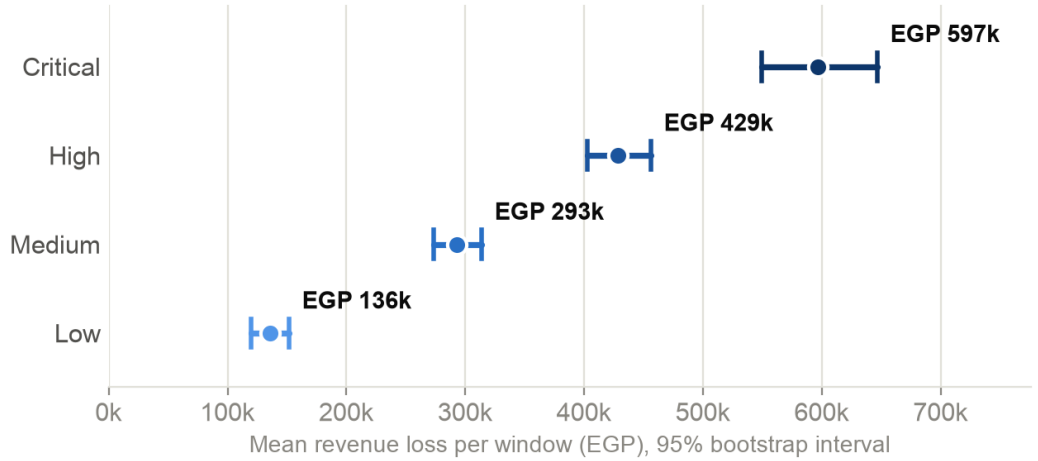



*Figure 12. Mean revenue loss by criticality with 95% bootstrap intervals; every step is clear of the next.*

SECTION 17

# Asset classes, environment, and what does not matter

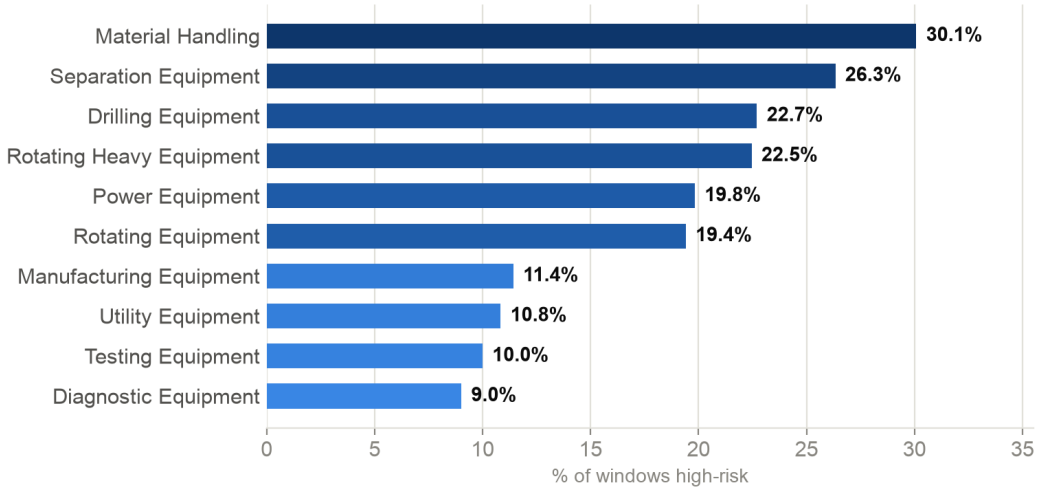



*Figure 13. Share of windows that are high-risk, by asset class.*

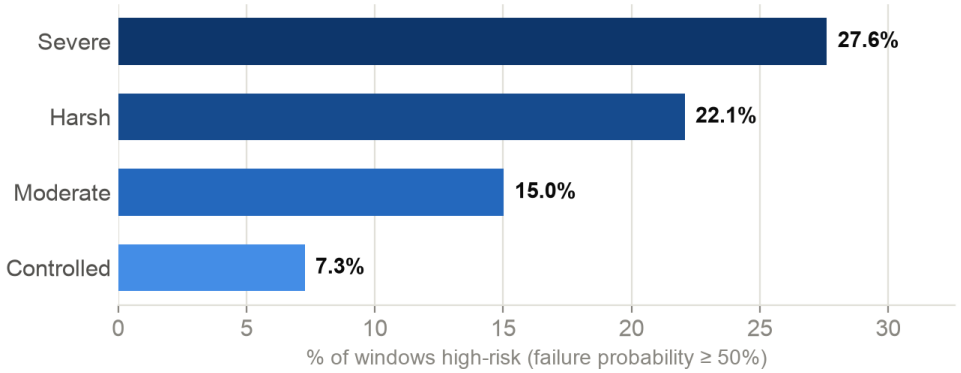



*Figure 14. Share of windows that are high-risk, by environment severity.*

## Tested and found to carry no signal

| Dimension | Downtime p | Loss p | Failure-prob. p | Verdict |
|---|---|---|---|---|
| Manufacturer Group | 0.57 | 0.75 | 0.90 | No difference |
| Operating Shift | 0.49 | 0.01 | 0.93 | No difference |

No OEM switch, shift restructuring or site-versus-site campaign is supported by this data. Environment is a real risk factor but a condition to plan around, not a lever.

SECTION 18

# Finding 4: the escalation gap

Executive Intervention Priority is a fixed banding of the Maintenance Priority Index (Section 6), an index that correlates only 0.75 with failure probability. Of 960 high-risk windows, 829 are not escalated, carrying EGP 723.3M of exposure (32.6% of the total). Of the 500 largest-loss windows, P1/P2 cover 26.5% of the loss.

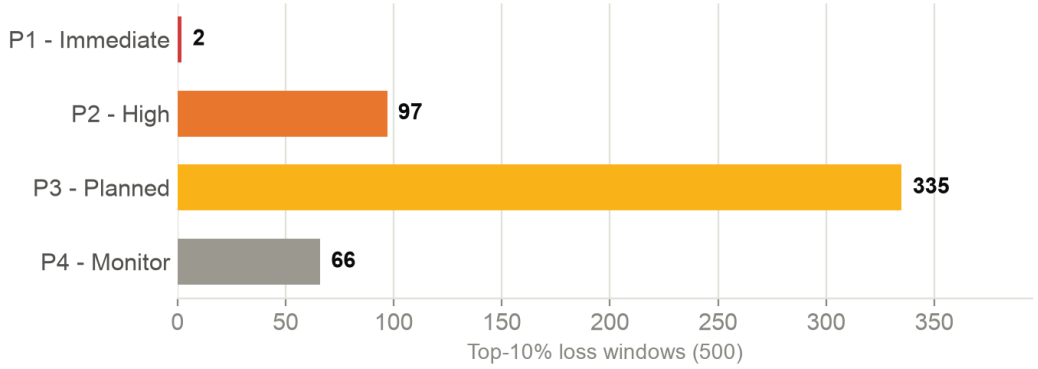



*Figure 15. The 500 largest-loss windows by the flag they received.*

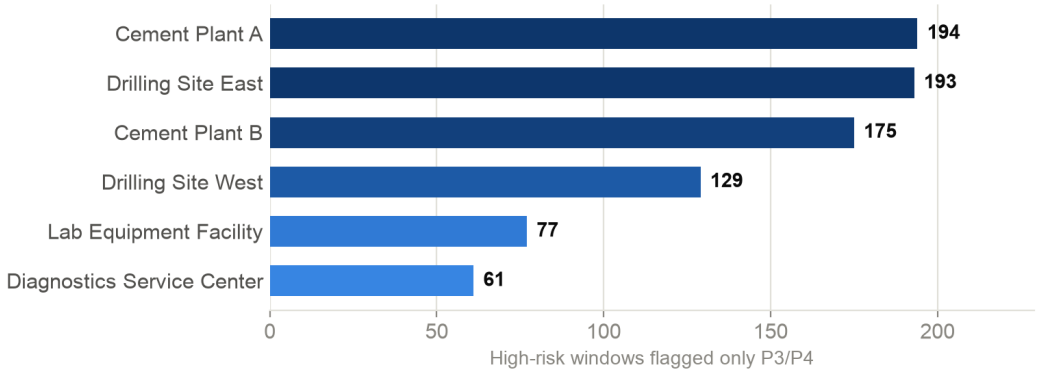



*Figure 16. Unescalated high-risk windows by site.*

**RECOMMENDED RULE**

Add a second trigger: failure probability ≥ 50% and criticality High or Critical. On this file it adds **433** windows and **EGP 434.3M** of exposure (60% of the gap), raising escalated windows from 134 to 567. Confirm crew capacity before switching it on.

SECTION 19

# The financial view

**EGP 1.88B**
revenue loss, all windows

**EGP 126.4M**
maintenance spend

**EGP 16.9M**
spare-parts spend

**14.9x**
loss per EGP of maintenance

Maintenance spend barely moves with criticality while loss rises steeply: the budget is not following the risk.

| Criticality | Windows | Maint. cost / window | Revenue loss / window | Loss per EGP spent |
|---|---|---|---|---|
| Critical | 880 | EGP 26.6k | EGP 596.5k | 22.4x |
| High | 1,789 | EGP 25.5k | EGP 428.6k | 16.8x |
| Medium | 1,724 | EGP 25.1k | EGP 292.7k | 11.6x |
| Low | 607 | EGP 23.1k | EGP 135.5k | 5.9x |

## Sizing the levers

| Scenario | Downtime avoided | Share of downtime | Maintenance spend avoided |
|---|---|---|---|
| Reactive windows run at the preventive average | 8,621 h | 16% | EGP 10.0M |
| Shortage windows run at the available-stock average | 5,800 h | 11% | EGP 8.2M |

**ESTIMATED, NOT MEASURED**

These are upper bounds, not forecasts. They assume each association is fully causal; reactive and shortage windows also start in worse health, and the two scenarios overlap where reactive windows lack parts. A controlled pilot measures the true share.

SECTION 20

# The significance register

All 33 tests, ranked by effect size. A difference counts only if ANOVA and Kruskal-Wallis both clear p < 0.0015. 26 survive.

| Dimension | Measure | F | p (ANOVA) | H | p (K-W) | $\eta^2$ | Highest group | Verdict |
|---|---|---|---|---|---|---|---|---|
| Failure Mode Group | Failure prob. | 971.6 | <1e-300 | 1837.7 | <1e-300 | 0.3684 | Bearing / Vibration | **Real, Large** |
| Maintenance Strategy | Failure prob. | 396.0 | 9e-231 | 806.0 | 2e-174 | 0.1921 | Reactive | **Real, Large** |
| Spare Parts Status | Failure prob. | 410.4 | 1e-165 | 678.4 | 5e-148 | 0.1411 | Shortage | **Real, Large** |
| Failure Mode Group | Downtime | 210.5 | 1e-128 | 596.0 | 7e-129 | 0.1122 | Bearing / Vibration | **Real, Medium** |
| Failure Mode Group | Revenue loss | 169.3 | 2e-104 | 500.7 | 3e-108 | 0.0923 | Bearing / Vibration | **Real, Medium** |
| Spare Parts Status | Downtime | 249.4 | 6e-104 | 528.5 | 2e-115 | 0.0908 | Shortage | **Real, Medium** |
| Maintenance Strategy | Downtime | 136.7 | 4e-85 | 369.6 | 9e-80 | 0.0758 | Reactive | **Real, Medium** |
| Environment Severity | Failure prob. | 119.8 | 6e-75 | 407.2 | 6e-88 | 0.0671 | Severe | **Real, Medium** |
| Spare Parts Status | Revenue loss | 174.0 | 9e-74 | 345.7 | 9e-76 | 0.0651 | Shortage | **Real, Medium** |
| Criticality Class | Revenue loss | 113.7 | 3e-71 | 385.4 | 3e-83 | 0.0639 | Critical | **Real, Medium** |
| Site | Failure prob. | 49.8 | 2e-50 | 283.3 | 4e-59 | 0.0475 | Cement Plant B | **Real, Small** |
| Industry Context | Failure prob. | 82.2 | 7e-52 | 281.1 | 1e-60 | 0.0470 | Cement Manufacturing | **Real, Small** |
| Maintenance Strategy | Revenue loss | 81.9 | 1e-51 | 263.3 | 9e-57 | 0.0469 | Reactive | **Real, Small** |
| Asset Class | Failure prob. | 20.2 | 9e-34 | 208.4 | 6e-40 | 0.0352 | Material Handling | **Real, Small** |
| Technician Skill Level | Failure prob. | 42.9 | 2e-27 | 129.8 | 6e-28 | 0.0251 | Basic | **Real, Small** |
| Technician Skill Level | Downtime | 41.4 | 2e-26 | 110.1 | 1e-23 | 0.0243 | Basic | **Real, Small** |
| Environment Severity | Downtime | 32.0 | 2e-20 | 119.3 | 1e-25 | 0.0188 | Severe | **Real, Small** |
| Technician Skill Level | Revenue loss | 28.1 | 5e-18 | 91.3 | 1e-19 | 0.0166 | Basic | **Real, Small** |
| Environment Severity | Revenue loss | 27.0 | 3e-17 | 103.0 | 3e-22 | 0.0159 | Severe | **Real, Small** |
| Site | Downtime | 15.2 | 7e-15 | 104.7 | 5e-21 | 0.0150 | Cement Plant B | **Real, Small** |
| Industry Context | Downtime | 25.1 | 4e-16 | 102.8 | 4e-22 | 0.0148 | Cement Manufacturing | **Real, Small** |
| Site | Revenue loss | 13.3 | 7e-13 | 98.5 | 1e-19 | 0.0131 | Cement Plant A | **Real, Small** |
| Industry Context | Revenue loss | 22.1 | 3e-14 | 97.5 | 5e-21 | 0.0131 | Cement Manufacturing | **Real, Small** |
| Asset Class | Downtime | 6.7 | 1e-09 | 86.2 | 9e-15 | 0.0119 | Material Handling | **Real, Small** |
| Asset Class | Revenue loss | 6.1 | 2e-08 | 79.5 | 2e-13 | 0.0109 | Material Handling | **Real, Small** |
| Criticality Class | Failure prob. | 14.5 | 2e-09 | 48.1 | 2e-10 | 0.0086 | Critical | **Real But Negligible** |
| Criticality Class | Downtime | 3.5 | 0.015 | 21.9 | 7e-05 | 0.0021 | Critical | **Not Significant** |
| Operating Shift | Revenue loss | 4.3 | 0.013 | 0.3 | 0.843 | 0.0017 | Night | **Not Significant** |
| Manufacturer Group | Downtime | 0.8 | 0.567 | 3.9 | 0.564 | 0.0008 | OEM-A | **Not Significant** |
| Manufacturer Group | Revenue loss | 0.5 | 0.747 | 6.1 | 0.297 | 0.0005 | OEM-A | **Not Significant** |
| Manufacturer Group | Failure prob. | 0.3 | 0.899 | 1.0 | 0.963 | 0.0003 | OEM-E | **Not Significant** |
| Operating Shift | Downtime | 0.7 | 0.486 | 0.2 | 0.885 | 0.0003 | Night | **Not Significant** |
| Operating Shift | Failure prob. | 0.1 | 0.932 | 0.3 | 0.864 | 0.0000 | Evening | **Not Significant** |

SECTION 21

# Limitations and risks

| Limitation | Effect on conclusions | Mitigation |
|---|---|---|
| Observational data | Lever effects are associations; reactive and shortage windows start in worse health | Controlled pilot before full budget (Rec. 6) |
| Asset_ID not a stable key | No per-asset history, remaining life or replacement plan | Fix the key in the next export (Rec. 7) |
| Overlapping 30-day windows | Summed totals are not annual figures | All comparisons use means per window |
| Derived columns | Scores restate their inputs; exposure ≈ loss | Excluded from drivers; never added together |
| Capped indices | Efficiency index 53% at 100; PM compliance 16% at 100 | Read values at the cap as 'at least' |
| Rare categories | P1 (2 rows), three failure modes (27 rows) | Pooled; not analysed alone |
| Twelve months only | No multi-year seasonality can be ruled out | Re-run on the next year's export |
| Escalation rule workload | 433 extra escalations may exceed crew capacity | Capacity check before go-live (Rec. 5) |

## What may and may not be stated publicly

| Safe to state | Do not state |
|---|---|
| Lever differences with their measured sizes | That a lever causes the reduction, until a pilot measures it |
| Criticality and environment as real risk factors | That any site is better or worse managed |
| Concentration figures and the escalation gap | Anything about manufacturer or shift quality |
| Means per 30-day window | Per-asset histories, remaining life by machine, annual totals |

SECTION 22

# Recommendations and implementation roadmap

The roadmap below is proposed, not measured: it sequences the seven recommendations so that each decision is taken with evidence from the step before it. Every KPI named is already on the live dashboard.

| Phase | Actions | KPI to watch (dashboard page) | Decision gate |
|---|---|---|---|
| 0–30 days | Approve pilot design; capacity check for the second trigger; request a stable Asset_ID | Escalation gap (Risk Watchlist) | Pilot site and control site chosen |
| 30–90 days | Convert the pilot site's reactive critical/high assets to planned; switch on the second trigger if capacity allows; stock shortage-prone spares at the pilot site | Downtime by strategy and spares (Levers); escalated count (Overview) | Pilot downtime vs control, with intervals |
| 90–180 days | Scale what the pilot proved; manage to PM ≥ 80% and backlog < 14 days fleet-wide; pair basic crews with experts on high-risk work | PM deciles and thresholds (Levers); high-risk share (Overview) | Budget re-allocation by measured effect |
| 180+ days | Re-run the analysis on the new export with a stable Asset_ID; add per-asset remaining-life and replacement views | All; Data & Method re-baselined | Next annual plan |

## Decisions needed this quarter

- **Approve the reactive-to-planned pilot**, one site with a matched control, measured on the live dashboard.
- **Approve the second escalation trigger**, subject to a capacity check for 567 escalated windows.
- **Request a stable Asset_ID** in the next export, so the next review can plan per asset.

APPENDIX

# Reproducibility and the live dashboard

| Artifact | Command / location | Notes |
|---|---|---|
| Analysis | python3 02_Analysis/analyze_pmaris.py | Standard library only, about 90 s; writes findings.json |
| Figures | python make_figures.py | matplotlib; writes 04_Deliverables/figures/ |
| This report | python make_full_report.py | reportlab |
| 10-page assessment | python make_report.py | reportlab |
| 30-slide deck | python make_deck.py | python-pptx, native charts, speaker notes |
| Executive presentation | python make_exec_deck.py | 15-slide subset of the full deck |
| Live dashboard | python3 watcher.py and python3 -m http.server 8000 in 03_Dashboard | Open http://localhost:8000/index.html |

The six live dashboard pages (Overview, Explorer, Levers, Sites & Equipment, Risk Watchlist, Data & Method) are described in the project README.

## Verification performed

- All six pages rendered headless in light, dark and 390 px phone width: zero console errors, no horizontal scroll.
- Live refresh: ten timed edits to the source CSV, 376 ms mean and 621 ms maximum from save to screen; source restored byte-identical.
- A half-written row was appended mid-test; the watcher skipped it and kept polling.
- Key figures cross-checked between this report, the 10-page assessment, both decks and the dashboard.